

Programme	B. Sc. Mathematics Honours			
Course Code	MAT7CJ402			
Course Title	GENERAL TOPOLOGY			
Type of Course	Major			
Semester	VII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	3	2	75
Pre-requisites	1. Mathematical Logic and necessary exposure to set theory. 2. Basic Calculus 3. Real Analysis I, Real Analysis II			
Course Summary	The subject of general topology is introduced with motivations from the theory of real functions and of metric spaces. Basic concepts like open and closed sets, interiors, closures, boundaries, neighbourhoods, bases and sub-bases are introduced. After a discussion of continuity and related topics, the universal properties of strong and weak topologies are discussed. Compactness, connectedness, and various countability axioms are studied in some detail. After a detailed study of the hierarchy of separation axioms and their interplay with other properties such as compactness, the course concludes with a presentation of the famous Urysohn & Tietze characterisations of normality.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define and classify topological spaces, bases, and subspaces, and apply these concepts to identify examples of different topological structures.	Ap	C	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
CO2	Analyse and evaluate closed sets, interior points, and accumulation points within topological spaces, and understand the concepts of continuity and related topological properties.	An	P	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
CO3	Synthesize the concepts of connectedness, separation axioms, and compactness to determine specific topological properties of spaces and analyse their applications in solving problems related to paths and separation.	E	C	Internal Exam/Assignment/Seminar/ Viva/Report/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Introduction to General Topology, K. D. Joshi,, New Age International Publishers, 1983.			
Module	Unit	Content	Hrs (45+30)	External Marks (70)
I	Topological Spaces		12	Min.15
	1	Chapter 4 – Section 1: Definition of Topological Space		
	2	Chapter 4 – Section 2: Examples of Topological Spaces		
	3	Chapter 4 – Section 3: Bases and Sub-bases – 3.1 to 3.7		
	4	Chapter 4 – Section 3: Bases and Sub-bases – 3.8 to 3.10		
	5	Chapter 4 – Section 4: Subspaces – 4.1 to 4.6		
II	Basic concepts		10	Min.15
	6	Chapter 5 – Section 1: Closed Sets and Closure (Proof of Theorem 1.5 is optional)		
	7	Chapter 5 – Section 2: Neighbourhoods, Interior and Accumulation Points – 2.1 to 2.8		
	8	Chapter 5 – Section 2: Neighbourhoods, Interior and Accumulation Points –2.9 to 2.10 and 2.13		
	9	Chapter 5 – Section 3: Continuity and Related Concepts – 3.1 to 3.6		
	10	Chapter 5 – Section 3: Continuity and Related Concepts – 3.7 to 3.11		
III	Spaces with special properties		12	Min.15
	11	Chapter 5 – Section 4: Making Functions Continuous, Quotient Spaces – 4.1 to 4.7		
	12	Chapter 5 – Making Functions Continuous, Quotient Spaces – 4.8 to 4.12		
	13	Chapter 6 – Section 1: Smallness Conditions on a Space – 1.1 to 1.9		
	14	Chapter 6 – Section 1: Smallness Conditions on a Space – 1.10 to 1.18		
	15	Chapter 6 – Section 2: Connectedness – 2.1 to 2.6 (Proof of Theorem 2.5 is optional)		
	16	Chapter 6 – Connectedness – 2.7 to 2.15		
IV	Separation axioms		11	Min.15
	17	Chapter 6 – Section 3: Local Connectedness and Paths – 3.1 to 3.8		
	18	Chapter 7 – Hierarchy of Separation Axioms - 1.1 to 1.6.		
	19	Chapter 7 – Hierarchy of Separation Axioms - 1.7 to 1.12		
	20	Chapter 7 – Hierarchy of Separation Axioms - 1.13 to 1.17		
	21	Chapter 7 – Section 2: Compactness and Separation Axioms - 2.1 to 2.6		

	22	Chapter 7 – Section 2: Compactness and Separation Axioms- 2.7 to 2.10		
V	Practicum:		30	-
Practicum	The goal is for the students to learn the following selected topics in 10 practicum sessions of hours each via self-study and group activities. The lecturer may assist by running group discussions, supervising class seminars and referring library books for self-study and note preparation.			
1	Chapter 1 - Logical Warm-up			
2	Chapter 2 – Preliminaries			
3	Chapter 3 – Motivation for Topology			
4	Chapter 6 - Connectedness: Theorem 2.5 and its proof			
5	Chapter 6 - Local connectedness and Paths - 3.9 to 3.11			
6	Chapter 7 - Compactness and Separation Axioms - 2.11 to 2.16			
7	Chapter 7 – Section 3: Urysohn Characterisation of Normality -3.1 to 3.4			
8	Chapter 7 – Section 3: Urysohn Characterisation of Normality - 3.5 to 3.6			
9	Chapter 7 –Section 4: Tietze Characterisation of Normality - 4.1 to 4.5			
10	Chapter 7 –Section 4: Tietze Characterisation of Normality - 4.6 to 4.8			
References				
1. Topology, J. R. Munkres, Prentice Hall of India, 2000.				
2. General Topology, S. Willard, Addison Wesley Pub. Company, 1976.				
3. General Topology, J. L. Kelley, D. van Nostrand, 1955.				
4. Introduction to Topology and Modern Analysis, G. F. Simmons, McGraw-Hill, 1963.				
5. Topology, James Dugundji, Prentice Hall of India, 1975.				

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	2	0	3	0	3	0	3	0	0
CO 2	3	2	2	1	3	0	3	0	3	0	0
CO 3	3	3	3	2	3	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Report
- Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/ Report	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓